Fixed-input Diophantine equations for the consistency of ZFC

A 70-operation polynomial and an eight-equation system

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Abstract

We specify an ordinary integer-coefficient polynomial in sixteen strictly positive unknowns whose absence of solutions is provably equivalent in ZFC to the syntactic consistency of ZFC. Its evaluation uses 70 binary arithmetic operations, comprising 40 multiplications and 30 additions or subtractions, when its five fixed integer coefficients are supplied as constants. Its exact total degree is 169. A 71-operation variant has degree 156. A more transparent alternative is an eight-equation system in seventeen positive unknowns, evaluated by 61 arithmetic operations; its sum of squared residuals has degree 56 and an 84-operation evaluator.

The main simplification is specific to a fixed input. The repository's universal ordinary-input construction needs a separate Pell block to encode the input marker. Here that marker is a fixed coefficient, so the block can be removed. A second reduction permits a signed repunit condition and proves its sign only after recovering a power of two. Reusing this not-yet-verified repunit saves a further operation when the first scale is changed from wq to wq^2.

The coefficients are fully determined by the literal K215 transition table, an explicit 435-state normalization, a finite window predicate, and a finite arithmetic recipe. Their enormous binary expansions and the complete window list are not materialized. This is an ordinary mathematical proof, with reproducible circuit and finite-controller checks, building on the pinned repository's Pell, compiler, and consistency-calibration theorems. It is not a newly kernel-checked end-to-end theorem, an internal ZFC proof object, or a claim of global minimality.

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1 The precise result and the meaning of simplicity

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$. Let Con(ZFC) mean that the fixed ordinary first-order axiom system for ZFC has no finite derivation of falsity. The derivability convention is the one calibrated by the repository's K215 endpoint. Equivalent effective presentations of the same usual theory can be substituted using the usual formalized proof translations.

Theorem 1.1 (Main result)**.** *The coefficient recipe in Section 9 determines five specific positive integers β, K, M, N, W. With these constants, the polynomial $\mathcal{P}_{70}$ defined in Section 2 satisfies*

$$\mathrm{ZFC} \vdash \quad (\exists \mathbf{x} \in \mathbb{Z}_{>0}^{16} \quad \mathcal{P}_{70}(\mathbf{x}) = 0) \iff \neg\,\mathrm{Con(ZFC)}.$$

Consequently ZFC proves

$$\boxed{(\nexists \mathbf{x} \in \mathbb{Z}_{>0}^{16} \quad \mathcal{P}_{70}(\mathbf{x}) = 0) \iff \mathrm{Con(ZFC)}.}$$

No consistency premise is used to prove either equivalence.

The reference to ZFC provability means that the mathematical argument can be carried out in ordinary ZFC. It does not assert that this archive contains a finite string in a chosen ZFC proof language that has been mechanically checked as that derivation. Section 12 separates this claim from the repository's external Lean and Rocq theorems.

There is no invariant, universally agreed meaning of "the simplest Diophantine equation." Degree, variable count, coefficient height, expanded monomial count, arithmetic-circuit size, and proof length are different measures. This note makes its principal comparison using the repository's arithmetic model: each binary $+$, $-$, or $\times$ costs one operation; reuse of an already computed expression and access to a fixed integer cost zero. Multiplication by a fixed coefficient still costs one multiplication. Equality tests are listed separately for a system. The finalizer is charged when a system becomes one polynomial.

Construction	Positive unknowns	Equations	Operations	Degree
Main polynomial $\mathcal{P}_{70}$	16	1	$40M + 30A = 70$	169
Lower-degree variant $\mathcal{P}_{71}$	16	1	$40M + 31A = 71$	156
Budget-square variant $\mathcal{P}_{73}$	16	1	$41M + 32A = 73$	157
Transparent system	17	8	$33M + 28A = 61$	≤ 28
Its sum of squares	17	1	$41M + 43A = 84$	56
Inherited universal polynomial	18 + input	1	$47M + 37A = 84$	187

The last row concerns a uniform positive input and has a different contract. Our construction specializes the input and adds its marker as a fixed coefficient. Thus 70 is a fixed-instance improvement, not a new uniform universal-operation bound. None of these numbers is a lower bound over all possible Diophantine descriptions. The very large coefficient height is reported explicitly rather than included implicitly in the operation count.

2 The polynomial, with every variable and coefficient specified

The sixteen unknowns are

$$\boxed{J, F, \alpha, t, f, h, i, T, s, w, \tau, \eta, \zeta, y, Z, \gamma \in \mathbb{Z}_{>0}.}$$

The letter T is an auxiliary quotient and t is a transport quotient. The uppercase Z is a packed word; it is unrelated to the ring $\mathbb{Z}$. All expressions introduced next are polynomial abbreviations, not additional unknowns or uncharged arithmetic primitives.

First define the geometry and scales:

$$C = Z + W, \qquad u = C + \alpha, \qquad q = u + F, \tag{1}$$
$$v = u - Z, \qquad P = \beta J, \qquad X = wq^2,$$
$$Y = sq^3, \qquad E = XY, \qquad k = \eta + \zeta, \tag{2}$$
$$c = kY + \eta, \qquad a = E + Y, \qquad H = 4a + 3,$$
$$\Delta = a^2 + H = (a+1)(a+3), \qquad D = X + ac + \gamma H, \qquad L = E(kY). \tag{3}$$

Next define the packed index and auxiliary expressions:

$$R = (Pu + v)(q^2 - 1) + (M + qN)J, \tag{4}$$
$$V = c(Tf - 1) - Rf^2, \tag{5}$$
$$A_* = (i\Delta c^2)^2. \tag{6}$$

The seven factors are

$$\begin{aligned} U_0 &= q - P, \\ U_1 &= \tau^2 - L(L + k), \\ U_2 &= D^2 - \Delta c^2, \\ U_3 &= A_*(V^2 - y^2) + y^2, \\ U_4 &= k - hE - R, \\ U_5 &= (K + w)C + u - tP, \\ U_6 &= \Delta f^2 - A_*. \end{aligned} \tag{7}$$

The promised equation is

$$\boxed{\mathcal{P}_{70} = U_0U_1U_2U_3U_4U_5U_6 - \Delta = 0.} \tag{8}$$

This is one ordinary polynomial equation with integer coefficients. All exponents of unknowns in its definition are fixed small integers. In particular, the coefficient $W = 2^{b+2d}$ introduced below is a fixed integer; there is no existential exponent, computation-length parameter, hidden bounded universal quantifier, or oracle in (8). Appendix A gives the complete 70-row straight-line evaluator.

2.1 The five coefficients

The exact compiler recipe produces integers $b, L_0, d, B, D_C, D_R, M_C, M_F$, where $d = bL_0$ and $B = 2^d$. The five ports used by the polynomial are

$$\boxed{\beta = B - 1, \quad K = D_C + BD_R, \quad M = M_C, \quad N = M_F + B - 1, \quad W = 2^{b+2d}.} \tag{9}$$

The M_F in this formula is the native verification mask. The polynomial's N is its *shifted* source value. Omitting this shift gives a different and unjustified equation.

For the explicit K215 instance supplied here,

$$b = 5^{44}, \qquad L_0 = 5^{45}, \qquad d = 5^{89}, \qquad B = 2^{5^{89}}, \qquad W = 2^{5^{44}+2\cdot 5^{89}}.$$

The other coefficients are fixed finite sums of powers of 2^b, whose terms are specified by the literal controller and window predicate. These are exact descriptions of integers, not parameters waiting to be selected by a halting or consistency decision. Section 9 and `artifacts/compiler/COEFFICIENTS.md` give the full recipe.

3 What is inherited, and what changes here

The general connection between computably enumerable sets and Diophantine equations is classical. Jones's explicit universal constructions and the DPRM theorem provide the historical framework [1, 2]. Explicit provability encodings also predate this work [3], and full formal DPRM developments are available [4]. The contribution here is a specific reduction and instantiation of the much more recent constructions already present in ProveIt.

All repository dependencies are pinned at `cdd582bec13f908df3711fa9e05d1ddac368a5db`. The argument uses three substantive inherited components.

1. **Consistency calibration.** The exact K215 machine halts from blank tape exactly when there is a finite ZFC refutation. The ordinary proof says this equivalence is formalizable in ZFC; external Lean and Rocq proof lanes establish their corresponding endpoint statements [5].
2. **Pell decoding.** The strong half-binomial kernel, its auxiliary rank and signed step-down lemmas, and the asymmetric-scale argument recover the actual main Pell index and a power of two from polynomial norm and ratio conditions [6, 7, 8].
3. **The fixed helical compiler.** The modified75 layout turns finite local windows into fixed coefficients, with shifted masks, synchronization, a unique start/end marker argument, and a five-adic dummy adjustment for completeness [9, 10].

We do not replace any of these by numerical samples. The new argument removes the input-Pell block, proves a weaker initial repunit condition sufficient, reuses its multiple before the sign is known, and supplies an explicit controller instance. The proof below records precisely where the inherited local lemmas are used. It does not invoke the whole universal polynomial theorem with one of its hypotheses silently deleted.

4 Why the input-Pell block can be deleted

In the modified75 compiler, ordinary input $x > 0$ is represented by a single End bit at the integer

$$W_x = 2^{2dx+b}.$$

A four-equation Pell block is needed to enforce this expression while x varies. We fix ordinary input $x_* = 1$, so the marker is the coefficient $W = W_{x_*}$ in (9). The identity $C = Z + W$ inserts that marker directly. No variable exponent remains to be enforced.

Lemma 4.1 (Fixed-marker specialization)**.** *For coefficients from the modified75 compiler, retain its full native kernel, packing, and transport conditions, set $C = Z + 2^{b+2d}$, and retain a positive bound $C < q$. Its soundness argument decodes ordinary input 1 without the four input-Pell equations. Every accepting computation at input 1 still has positive witnesses for these retained conditions.*

Proof. The native kernel and packing argument recover $q = B^m$ and the shifted mask conditions. Since $W < C < q$ and $2^b < B$, necessarily $m > 2$. The native Z word has its End selector forbidden,

so adding $W = 2^b B^2$ has no carry and inserts exactly one End at cell distance 2. The shifted low mask supplies Start at the origin. The existing marker bijection then makes Start unique and identifies these two markers with the same initialization run. The run has three I cells and its Q head, hence four unary 1 symbols. This is ordinary input 1 in the modified75 convention. Conversely, every old complete witness has exactly this marker; forgetting its five private input-Pell coordinates and treating W as a coefficient preserves all retained equations. The canonical padding theorem supplies any stronger positive budget used below. □

The explicit normalizer in Section 9.1 erases its unary initialization and runs K215 on a blank tape. It therefore has the same halting behavior for every allowed unary length. Nevertheless, the marker proof is still needed: it guarantees a genuine initialized computation rather than an unrelated periodic component.

5 Soundness of the 70-operation polynomial

Fix the five actual coefficients and a positive tuple satisfying (8). We show that the tuple decodes an accepting computation. This direction is where false solutions must be excluded.

5.1 Cancel the positive discriminant before treating factors as units

All definitions of a are positive, so $\Delta = (a+1)(a+3) > 0$ before imposing any equation. Define

$$S = f^2 - \Delta i^2 c^4.$$

Then $U_6 = \Delta S$, and the exact polynomial identity is

$$\mathcal{P}_{70} = \Delta\big(U_0U_1U_2U_3U_4U_5S - 1\big). \tag{10}$$

Consequently the seven integer factors inside the parentheses have product 1, and each is 1 or -1. Inferring unit factors directly from a product equal to Δ would be invalid; the cancellation in (10) is essential.

Lemma 5.1 (The four norm signs)**.** *At a positive zero, $U_1 = U_2 = U_3 = S = 1$.*

Proof. Put $V_0 = XY^2$. The identity $L(L+k) = V_0(V_0+1)k^2$ shows that U_1 is a Pell norm. For $V_0 > 1$, the equation $r^2 - V_0(V_0+1)\ell^2 = -1$ has no solution with $\ell > 0$. Indeed, such a solution gives $V_0\ell < |r| < (V_0 + \frac{1}{2})\ell$ and hence the smaller positive integer

$$\ell' = (2V_0+1)\ell - 2|r|, \qquad 0 < \ell' < \ell.$$

Multiplying by the norm-one unit $2V_0+1-2\sqrt{V_0(V_0+1)}$ produces another norm-minus-one solution with this coefficient, contradicting minimality. Thus $U_1 = 1$.

The discriminant $\Delta = a^2 + 4a + 3$ is 0 or 3 modulo 4. A norm $r^2 - \Delta z^2$ therefore cannot be -1 modulo 4. This proves $U_2 = S = 1$. Finally, A_* is a square. Modulo 4, the expression $A_*V^2 - (A_*-1)y^2$ is a square whether A_* is 0 or 1 modulo 4. It cannot be -1, so $U_3 = 1$ as well. □

Write

$$U_0 = \epsilon, \qquad U_4 = \iota, \qquad U_5 = \nu, \qquad \epsilon, \iota, \nu \in \{1, -1\}, \quad \epsilon\iota\nu = 1.$$

We will establish their positive signs in the order ι, ϵ, ν. In particular the exact repunit relation is not available yet.

5.2 Bounds valid for both repunit signs

From the definitions, $C = Z + W > 0$, $q = C + F + \alpha$, and $F + Z < q$. Also $q > W \geq 2B^2$, so certainly $q \geq 16$. Since $\epsilon = q - \beta J$, we have $\beta J = q - \epsilon$ and

$$J \leq (q+1)/\beta. \tag{11}$$

The compiler masks satisfy

$$0 < M < \beta, \qquad 0 < M_F < \beta, \qquad N = M_F + \beta < 2\beta.$$

Let $G = q^2 - qF - Z$. The positive budget gives

$$2q - 1 \leq G \leq q^2 - q - 1.$$

Crucially, the actually computed first factor in (4) is

$$Pu + v = G + (1 - \epsilon)(q - F). \tag{12}$$

It is not permissible to replace it by G before proving $\epsilon = 1$. Since $1 \leq q - F \leq q - 1$, both signs imply

$$2q - 1 \leq Pu + v \leq q^2 + q - 3.$$

Using (11) in the mask term yields

$$R > (2q-1)(q^2-1) > 3q + 1, \tag{13}$$

$$\begin{aligned} R &< (q^2 + q - 3)(q^2 - 1) + (1 + 2q)(q + 1) \\ &= q^4 + q^3 - 2q^2 + 2q + 4 < q^4 + q^3. \end{aligned} \tag{14}$$

The revised scale $X = wq^2$, $Y = sq^3$ now supplies

$$E \geq q^5 > R + 5, \qquad a = E + Y > R + 2. \tag{15}$$

This extra margin is exactly why the repunit can be reused while its sign is still unknown. The former asymmetric scale $X = wq$ would not supply this bound for the negative branch of (12).

5.3 Restore the auxiliary congruence witnesses

The equality $S = 1$ gives

$$f^2 = 1 + \Delta i^2 c^4. \tag{16}$$

We have $c > 2$, hence $f > c^2 > 2c$. Since $R < a < \Delta$,

$$\begin{aligned} A_* - (Rf^2 + c) &= (\Delta - R)f^2 - \Delta - c \\ &\geq f^2 - \Delta - c = 1 + \Delta(i^2c^4 - 1) - c > 0. \end{aligned} \tag{17}$$

The auxiliary norm is

$$A_* V^2 - (A_* - 1)y^2 = 1.$$

It excludes $V = 0$. If $v_0 = |V| > 1$, then $v_0^2 - 1 = (A_* - 1)(y^2 - v_0^2)$ and $y \geq v_0 + 1$. It follows that $v_0 \geq 2A_* - 1$. But $V = c(Tf - 1) - Rf^2 > -Rf^2 - c$, so (17) excludes every negative value with $|V| > 1$. The remaining possibilities $V = \pm 1$ are impossible because $V \equiv -c \pmod f$ and $2 < c < f/2$. Thus $V > 0$.

Define the proof-only integers

$$o = cT - Rf, \qquad j = Tf - R\Delta i^2 c^3 - 1. \tag{18}$$

Then $of = V + c > 0$, and (16) gives $jc = V + R > 0$. Thus $o, j > 0$ and

$$V = of - c = jc - R. \tag{19}$$

The positivity is proved before applying a Pell-index theorem; it is not an assumption about the omitted coordinates.

5.4 The actual Pell index and the sign of the index factor

Write $A = a + 2$ and $P_0 = 2XY^2 + 1$. For $A \geq 2$ use

$$\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

The first and main positive norms give positive indices n, p with

$$k = 2\psi_{P_0}(n), \qquad \tau = \chi_{P_0}(n), \qquad c = \psi_A(p), \qquad D = \chi_A(p).$$

Here $P_0 > A$, $c > k$, and the two ratio slacks give

$$kY < c < k(Y + 1). \tag{20}$$

Parameter monotonicity implies $p > n$. Since $P_0 \equiv 1 \pmod{E}$,

$$2n \equiv R + \iota \pmod{E}, \qquad n \geq (R - 1)/2.$$

The lower bound on R makes $p \geq 25$. Standard Pell growth now gives $c > A\Delta^2$ and $c > 2p$. Also $k = R + \iota + hE$ and $Y \geq q^3$ give $c > 2R$.

At this point the inherited strong-rank and signed auxiliary step-down lemmas apply [6, 8]. To match their ordinary strong premise, put $i_{\text{old}} = \Delta i$; then

$$(i_{\text{old}} c^2)^2 = \Delta(f^2 - 1).$$

Their other hypotheses are the just-established positive o, j, V, the congruences (19), $c > A\Delta^2$, $c > 2p$, and $c > 2R$. They identify the main index as

$$p = R. \tag{21}$$

These are local applications of the rank and step-down lemmas. They do not use a decoded input, a typed word, the exact repunit equation, or the final universal-polynomial theorem.

The least positive representative of $R + \iota$ modulo E is itself. If $2n = R + \iota + vE$ with $v \geq 1$, (15) would give $2n > 2R$, contradicting $n < p = R$. Hence $2n = R + \iota$. If $\iota = -1$, then $R = 2n + 1$. Put $Q_2 = \chi_A(2) = 2A^2 - 1$. Since $Q_2 > P_0$ and $A > Y + 1$, Pell duplication gives

$$\psi_A(2n) = 2A\psi_{Q_2}(n) \geq Ak, \qquad c = \psi_A(2n + 1) > Ak > k(Y + 1),$$

contradicting (20). Therefore

$$\iota = 1, \qquad 2n = R + 1. \tag{22}$$

5.5 Recover a power of two before deciding the repunit sign

Set $r = n - 1$, so $R = 2r + 1$ and $r \geq 1$. The lower Pell-ratio estimate used in the asymmetric-scale proof [7] is

$$\frac{c}{k/2} > \xi, \qquad \xi = \frac{(X+1)^{2r}}{X^r}.$$

Its elementary hypotheses $X \geq 16$, $Y \geq 4096$, and $6XY^2 > a$ hold here. This lower estimate does not require a decoded word or an upper binomial-floor error estimate. Combining it with (20) yields

$$Y > \xi/2 - 1 > X^r/3, \qquad a = Y(X+1) > X^{r+1}/3 > 2 \cdot 4^r = 2^R.$$

The sequence $\chi_A(j) - a\psi_A(j)$ is congruent to 2^j modulo $H = 4a + 3$, as follows directly from its recurrence and initial values $1, 2$. The definition of D therefore gives $X \equiv 2^R \pmod{H}$. Both numbers lie strictly between 0 and a, so

$$X = 2^R. \tag{23}$$

Since $q \mid X$, we have $q = 2^\ell$ for some $\ell \geq 4$.

Lemma 5.2 (The negative repunit sign is impossible)**.** *Let $B = 2^d$ with $d \geq 3$. If q is a positive power of two and $q - (B-1)J \in \{1, -1\}$ for a positive integer J, then $q - (B-1)J = 1$.*

Proof. For the negative sign, $2^\ell \equiv -1 \pmod{2^d - 1}$. Writing $\ell = ad + r_0$ with $0 \leq r_0 < d$ gives divisibility of $2^{r_0} + 1$ by $2^d - 1$. This is impossible because

$$0 < 2^{r_0} + 1 \leq 2^{d-1} + 1 < 2^d - 1.$$

□

Thus $\epsilon = 1$, so $P = q - 1$. The product of the remaining unit signs now forces $\nu = 1$. Moreover $q \equiv 1 \pmod{2^d - 1}$ implies $d \mid \ell$ by the same reduction of the exponent; hence $q = B^m$.

5.6 Restore the native compiler equations and decode the computation

All preliminary errors have disappeared: (12) becomes $Pu + v = q^2 - qF - Z$, and the packing is exactly

$$R = (q^2 - Z - qF)(q^2 - 1) + (M_C + q(M_F + B - 1))J.$$

The original mask bounds now give $3q + 1 < R < q^4$. Equation (23) and $R > 3q > 3\ell$ imply $q^3 \mid X$, so

$$w_{\text{old}} = X/q^3 = w/q \in \mathbb{Z}_{>0}.$$

The proof-only substitution $i_{\text{old}} = \Delta i$ and the positive o, j from (18) restore the ordinary strong half-binomial kernel. Every restored definition and norm agrees exactly. If the preceding projection presentation keeps two positive slacks, they can also be restored: $c > A\Delta^2$ and the main norm imply $D > (a+1)c$, so $\gamma H = D - ac - X > c - X > A\Delta^2 - a > H$. Thus $\gamma \geq 2$; take $\rho = 1$ and $\sigma = \gamma - 1$. No extra equation is needed.

The transport condition $U_5 = 1$ reads

$$(K + w)C - F = (t - 1)(q - 1).$$

Consequently the positive integer

$$z = t - 1 + w(q+1)C$$

satisfies

$$(K + X)C = F + z(q - 1).$$

This is the native transport equation since $X = wq^2$. Also $C < q$, $Z < C$, and $C = Z + W$ hold by definition.

The inherited half-binomial population argument and modified75 synchronization now apply without alteration. They yield the typed cyclic word, the actual whole-cell temporal shift, and the local transition tests. Lemma 4.1 supplies the unique initialization markers at ordinary input 1. The finite helical soundness theorem yields a halting run of the explicit normalizer, hence a blank halting run of K215. This proves soundness.

6 Completeness and the ZFC equivalence

Suppose K215 halts on blank tape. The explicit normalized controller then halts on four unary 1 symbols. The inherited helical padding and modified75 compiler completeness theorems produce a positive native witness at ordinary input 1. The normalized strong and auxiliary-quotient refinements used in the repository's 84-operation polynomial supply all of the corresponding positive coordinates [11, 8].

Keep its $q, C, F, Z, J, k, c, a, D, f, h, i, T, s, \tau, \eta, \zeta, y$ and its main power X. Define the new positive quantities

$$W = 2^{b+2d}, \qquad \gamma = \rho + \sigma, \qquad \alpha = q - C - F, \qquad w = X/q^2.$$

The old stronger budget is $q = C + F + Z + \alpha_{\text{old}} + 2d$, so $\alpha = Z + \alpha_{\text{old}} + 2d > 0$. The old main power is divisible by q^3, hence the new w is a positive integer. The old input theorem gives exactly $C = Z + W$.

The native transport equation implies that

$$t = 1 + \frac{(K + w)C - F}{q - 1}$$

is an integer: $X \equiv w \pmod{q - 1}$. This quotient is greater than -1 because $0 < F < q$ and $(K + w)C > 0$. Therefore $t \geq 1$. This positivity check is necessary when changing the scale and the transport quotient; it does not follow merely from the positivity of the old quotient.

The new definitions recover every retained native expression. The factors $U_0, U_1, U_2, U_3, U_4, U_5$ equal 1 and $U_6 = \Delta$, so $\mathcal{P}_{70} = 0$. This proves completeness.

Combining the two directions gives

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{16}\ \mathcal{P}_{70}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \text{Halt}(K215, \text{blank}).$$

The inherited calibration $\text{Halt}(K215, \text{blank}) \leftrightarrow \neg\,\text{Con}(\text{ZFC})$ then proves Theorem 1.1. Every step is ordinary finite arithmetic, Pell theory, finite-computation coding, or an explicitly cited set-theoretic calibration. These arguments can be formalized in ZFC.

The construction does not prove that either side is false. If ZFC is consistent, the equation has no positive solution, and ZFC cannot prove that absence under the usual hypotheses of the second incompleteness theorem. Mere consistency is not being used to assert that ZFC also cannot prove the existence of a solution; that stronger independence statement requires an appropriate soundness hypothesis. Only the displayed equivalence is needed here.

7 An eight-equation alternative

For a more direct audit, use the same five coefficients and seventeen positive unknowns

$$J, F, \alpha, z, Z, f, h, i, j, o, s, w, \tau, \eta, \zeta, \gamma, y.$$

Define

$$\begin{aligned}
&C = Z + W, && q = C + F + \alpha, && X = wq^3, && Y = sq^3,\\
&E = XY, && k = \eta + \zeta, && c = kY + \eta, && a = E + Y,\\
&H = 4a + 3, && \Delta = a^2 + H, && D = X + ac + \gamma H, && L = E(kY),\\
&R = (q^2 - Z - qF)(q^2 - 1) + (M + qN)J,\\
&A_0 = \Delta(f^2 - 1), && V = of - c, && U = jc - R.
\end{aligned}$$

The eight equations are

$$\boxed{\begin{aligned}
\beta J &= q - 1,\\
(K + X)C &= F + z(q - 1),\\
\tau^2 - L(L + k) &= 1,\\
k - hE - R &= 1,\\
D^2 - \Delta c^2 &= 1,\\
(ic^2)^2 &= A_0,\\
A_0(V^2 - y^2) + y^2 &= 1,\\
U &= V.
\end{aligned}} \tag{24}$$

Here R is a computed expression, not a positive supplied coordinate. The budget $q = C + F + \alpha$ guarantees $F + Z < q$ and therefore the required positive range of R. Dropping F from this budget while also eliminating the supplied R would leave a gap in the proof.

The first norm identity

$$(E^2 + X)(kY)^2 = L(L + k)$$

holds as a polynomial identity. The strong equation makes $A_0 = (ic^2)^2$ at a simultaneous zero, and $U = V$ restores the original auxiliary argument. Thus (24) is exactly a fixed-marker projection of the ordinary symmetric half-binomial construction. The previous soundness and completeness proof applies, without the signed repunit optimization or the positive Bezout quotient elimination.

The emitted comparison circuit costs 61 operations. If $r_1, \dots, r_8$ are its literal left-minus-right residuals, then

$$\boxed{r_1^2 + \cdots + r_8^2 = 0} \tag{25}$$

is an equivalent single polynomial equation over either the positive or nonnegative reparameterized domain. Its finalizer costs eight subtractions, eight squares, and seven additions, giving $61 + 23 = 84$ operations.

The residual degree bounds, in the displayed order, are

$$1, 5, 26, 9, 22, 22, 28, 6.$$

The seventh has the unique degree-28 leading part $w^2 s^4 k^2 q^{18} f^2$, where the leading part of q is $Z + F + \alpha$. Hence the sum of squares has exact degree 56. The auxiliary coefficient $A_0 = \Delta(f^2 - 1)$ and argument $V = of - c$ are intentional: replacing them off the zero set by their equivalent high-degree expressions would worsen the polynomial degree without changing the simultaneous solutions.

8 The two nearby single-polynomial variants

The archive also contains two useful checkpoints.

For $\mathcal{P}_{71}$ keep the same positive variables and geometry $C = Z + W$, $q = C + F + \alpha$, but set $X = wq$ and use the exact expression $q - 1$ in packing and transport. Keep the signed factor $q - \beta J$. Its initial packed-index bound is

$$R < q^4 - q^3 + 4q + 2,$$

which already gives $E \geq q^4 > R + 5$. The same local argument recovers the dyadic q and then the positive repunit sign. Its evaluator has 71 rows and exact degree 156. It trades one additional subtraction for a smaller degree than the main construction.

For $\mathcal{P}_{73}$ impose $q = \beta J + 1$ by definition, set $C = Z + W$, take $X = wq$, and replace the signed repunit factor by

$$1 + (q - F - C - \alpha)^2.$$

At a zero this positive unit is 1, so the budget holds exactly. All other factors are the same normalized native expressions. This variant avoids the delayed repunit-sign proof; it has 73 operations and exact degree 157. The 70, 71, and 73 circuits represent the same acceptance condition but are not asserted to be identical polynomials off their zero sets or to have identical positive witness tuples.

9 An explicit finite program and coefficient recipe

9.1 Pinning the actual machine

The input is the exact released dense table `candidate-k215/k215.tm`, with SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

Its working states are $0, \ldots, 214$, its separate halt is 215, and its tape symbols are $0, 1$. It starts in state 0 at position 0 on the blank two-way tape. The archive includes this table and its upstream attribution.

The normalization is completely explicit. It has three loader states, one nonhalting trap, one absorbing halt, and two copies of each of the 215 working states, for 435 states in total. At physical position $-j$, $j \geq 0$, a pair stores original cells j and $-j - 1$. A side bit in the state selects the component being read. Four ordinary pair symbols and four origin-flagged pair symbols, together with loader symbols $1, 2$, give ten tape symbols. Blank is the ordinary pair $(0, 0)$.

Away from the flagged origin, moving further out on either side of the original tape moves left physically, while moving toward the origin moves right. At the flagged origin, crossing between original cells 0 and -1 changes the side tag and stays in place. The written component changes exactly as in K215 and the other component remains unchanged. The six-case motion table and every one of the 4,350 transitions are supplied in `artifacts/compiler/`.

The first transition, from the unary initialization, writes 2 and stays. The loader scans left erasing the remaining unary 1s, returns right across the erased zeros, replaces the 2 by a flagged blank pair, and enters the copy of original state 0 on side zero. It never visits a positive physical position. On the required four unary 1s this takes ten steps. Induction on the original transitions proves exact tape and state correspondence, including halt. The normalizer's full transition table is materialized and checked; this is not an unspecified universal-machine index.

9.2 The finite local-window alphabet

Use the repository's helical tableau predicate unchanged. A tile records a vertical-boundary bit, a horizontal-boundary bit, a tape symbol and optional state, and, on an initialization row, one of the phases L, I, Q, R. There are exactly

$$a_0 = 2 + 10(435 + 1) + 4 = 4366$$

tiles, with their precise order and local predicate supplied by `finite_window_constants.py`. An allowed 3×3 window is one whose central cross satisfies that predicate; its four corners are free. The ordered window list begins with the designated Start and End windows, then lists all other accepted nine-tuples in lexicographic tile order.

This definition entails at most 4366^9 tests of a fixed finite predicate. It is huge but finite and independent of whether K215 halts. A compressed exact count, with a disjoint-category proof, gives

$$k_0 = 113175624520853754856480928432$$

windows. The full list is not expanded in this archive. Its definition is nevertheless complete: there is no unprovided choice of a machine, transition, tile, window, index, or coefficient.

The local conditions ensure that a Start sits in a finite initialized rectangle in the periodic covering plane. The phase graph through Q forces vertical boundaries on both sides, horizontal boundaries propagate across the strip, and a next horizontal boundary is reached by vertical periodicity. Between them, the exact update and no-escape rules preserve the unique head. The final-row head is required to be halted. This is the inherited soundness argument that excludes spurious accepting cycles. Conversely every finite one-sided halting run fits into sufficiently large independently padded width and height, including the powers of five needed by the compiler.

9.3 The exact fixed integers

The full coefficient construction is reproduced in Appendix B and the companion `COEFFICIENTS.md`. It specifies every clause coefficient, native position, copy shift, mask bit, anchor, and optional correction by finite sums. Its exact interfaces are the reviewed `compile_windows` and modified75 `new_constants` recipe, written out in a standalone form; no historical module is imported at runtime. In particular:

1. Each window contributes its selector coefficient and the one-hot indicators of its nine tile entries. Finitely many coefficient-zero dummy clauses make room for the native copy positions.

2. Two copies of the center clauses, six horizontal and six temporal overlap families, and four separated anchors determine sparse polynomials D_C, D_R in $R_0 = 2^b$.

3. A single optional high monomial is added to D_C exactly when the specified finite sum has $2D_C - D_R = 0 \pmod 5$ before that correction. This bit is a *defined constant*, not a free coefficient or a consistency test. It ensures the five-adic dummy coefficient is a unit.

4. The native and verification masks are those of the modified75 construction. One ignored extra bit is freed in the native mask and the low verification bit 4 is added, retaining the two-bit global population adjustment.

5. The radix and cell length are the least powers of five meeting the specified carry and separation bounds. Their evaluated values are $b = 5^{44}$ and $L_0 = 5^{45}$. Thus $d = 5^{89}$.

The integer d itself is

$$161558713389263217748322010169914619837072677910327911376953125.$$

Even $B = 2^d$ consequently has $d + 1$ binary digits. The fact that these coefficient expansions are infeasible to print must not be confused with an unknown coefficient. The finite recipe defines them uniquely and proves their required properties without expanding them. The optional mod-5 correction and all window-dependent sums are retained symbolically.

The delivered controller JSON and coefficient specification are the actual instantiation artifacts. The archive does not claim to have evaluated every huge coefficient, enumerated every window, or exhibited a positive solution to the target equation.

10 Exact degree and circuit-size verification

The 70-row evaluator uses only binary additions, subtractions, and multiplications. A topological and liveness audit checks that every row is defined before use, every supplied variable and coefficient is live, and all rows lead to the single output. The count includes all six final multiplications and the final subtraction of Δ.

For a uniform degree calculation, put

$$Q = F + Z + \alpha, \qquad k_0 = \eta + \zeta.$$

These are the leading homogeneous parts of q and k. The exact degrees of $(U_1, U_2, U_0, U_3, U_4, U_5, U_6)$ are

$$24, 20, 1, 64, 8, 2, 50,$$

whose sum is 169. Cancellation in $D^2 - \Delta c^2$ must be accounted for; naively propagating gate degrees gives only an upper bound. The complete leading homogeneous form is

$$-8Q^{104}h\gamma i^4 k_0^{13} w^{13} s^{26} T^2 f^2 \, (Q - \beta J)(wZ - \beta Jt). \tag{26}$$

For example, the coefficient of

$$J^2 F^{104} h\gamma i^4 \eta^{13} w^{13} s^{26} T^2 f^2 t$$

is $-8\beta^2 \neq 0$. This proves exact degree 169 for every valid compiler instance, including the present one. Subtracting Δ, of degree 14, cannot affect this leading term.

The executable evidence includes exact integer evaluations of the complete circuits, checks of the scaled-output and packing identities, and independent dense univariate degree computations at two finite-field specializations. Those computations supplement the displayed symbolic proof; they are not the proof of the all-size consistency equivalence or a substitute for the uniform nonzero leading coefficient.

11 Nonnegative and unrestricted integer solutions

The domain in Theorem 1.1 is essential. For a version over $\mathbb{N}^{16}$, replace each positive unknown x_j by $u_j + 1$ and set

$$\mathcal{P}_{\mathbb{N}}(u_1, \ldots, u_{16}) = \mathcal{P}_{70}(u_1 + 1, \ldots, u_{16} + 1).$$

This is a concrete polynomial of the same degree 169, and its absence of nonnegative solutions is equivalent to Con(ZFC). The direct shift adds sixteen additions to the displayed evaluator.

For a polynomial with *unrestricted integer* unknowns, use four squares for each positive coordinate:

$$x_j = 1 + z_{j,1}^2 + z_{j,2}^2 + z_{j,3}^2 + z_{j,4}^2.$$

Lagrange's four-square theorem proves that every positive integer occurs. Hence the explicitly substituted polynomial

$$\mathcal{P}_{\mathbb{Z}}(\mathbf{z}) = \mathcal{P}_{70}\left(1 + \sum_{r=1}^{4} z_{1,r}^2, \ldots, 1 + \sum_{r=1}^{4} z_{16,r}^2\right) \tag{27}$$

has 64 integer unknowns, exact degree 338, and no integer solution exactly when ZFC is consistent. To see that the degree is exact, substitute the sums of four squares into the nonzero degree-169 leading form. Setting three coordinates in each group to zero leaves its substitution $x_j = z_{j,1}^2$, which is nonzero because distinct monomials remain distinct. Hence the degree-338 part does not cancel. The direct evaluator costs $70 + 16 \cdot 8 = 198$ operations. No minimization is claimed for this domain conversion.

Using the raw 16-variable polynomial over all of $\mathbb{Z}^{16}$ would be incorrect. In particular the proof uses positive Δ to cancel a factor, positive ratio slacks, and positive bounds on the packed word. The four-square substitution supplies those conditions polynomially. The archive includes the complete 86-row and 198-row evaluators in `artifacts/domain_variants.json`.

12 Logical and verification status

The exact Lean endpoint inspected in the pinned repository is

```
BlankHalts machine215 <-> BProv ZFCax_s [] Form.fBot
```

and its nonhalting statement is obtained by negating this equivalence. The repository records completed Lean and Rocq verification for the literal machine. This work reads those statements and retained records and checks the table identity; it does not report a new kernel rebuild.

The calibration passes through the compact numerical calculus C_{222}:

$$\mathrm{Halt}(K215, \mathrm{blank}) \longleftrightarrow C_{222}(1) \longleftrightarrow \mathrm{ZF} \vdash \bot \longleftrightarrow \mathrm{ZFC} \vdash \bot.$$

The last equivalence is the inherited relative-constructibility calibration. It is about finite refutability, not an assumption that a model of ZFC exists. The ordinary mathematical proof in the K215 source explicitly identifies its ZFC formalizability claim. An interpretation of Lean's entire hierarchy of universes inside ZFC is neither asserted nor needed here.

The present artifact has the following evidence boundaries.

Claim	Evidence supplied
Exact K215 identity	Pinned table and SHA-256; inherited endpoint statements.
One-sided normalization	Explicit table, tape invariant proof, exhaustive valid folded-row checks, and loader checks.
Finite coefficient specification	Exact local predicate, ordered finite window definition, clause/mask formulas, scalar layout metadata.
Fixed-input reduction	Ordinary proof in Sections 4–7, using stated inherited Pell and compiler lemmas.
Arithmetic count and degree	Complete acyclic circuits, independent formula checks, liveness/count receipts, symbolic leading forms.
New formal end-to-end theorem	Not supplied; a proposed formalization task.
Expanded giant coefficients or target witness	Not supplied and not used as evidence.

13 Reproduction and repository integration

The archive is self-contained for its newly supplied Python checks; only the standard library is required. From its root run

```
python3 artifacts/fixed_input.py --output artifacts/fixed_input.json
python3 artifacts/domain_variants.py --output artifacts/domain_variants.json
python3 artifacts/compiler/k215_one_sided.py
python3 artifacts/compiler/finite_window_constants.py
python3 artifacts/review/review_fixed70.py
```

The root wrapper `python3 verify.py` runs these commands in order; the source manifest records the pinned upstream paths and hashes. The routines perform finite structural and algebraic checks. They do not run K215 until halt, enumerate its proof search, allocate the full window list, or expand the giant coefficients.

Compile this article with two runs of pdfLaTeX from the `report/` directory. The complete gate appendix is a small generated TeX include, and is shipped alongside the main source. The root README gives the exact commands and the generated JSON interfaces.

Suggested integration is a new directory under `Computability/HilbertTenthProblem/Papers/research-wip/` for the article and artifacts, with cross-links to the pinned K215 endpoint and the modified75 compiler. The existing universal84 and certificate74 records should remain unchanged: the new statements have fixed-input contracts. This delivery does not push or modify the user's repository.

14 Further research questions

1. **Kernel-check the fixed-input adapter.** Formalize the exact eight-equation projection first, then the auxiliary quotient and signed repunit reductions. The main new proof obligations are bounded polynomial identities, positivity, and Lemma 5.2; the existing K215 endpoint can remain an imported theorem.

2. **Replace the large finite window enumeration.** The present recipe is fully explicit but encodes many windows that never occur in a genuine initialized run. A reachability-preserving sparse atlas could reduce the coefficient description and height substantially while leaving the 16-variable arithmetic shell unchanged.

3. **Compile the numerical proof calculus directly.** The repository's C_{222} calculus has polynomial formula constructors and a short enumerator. A direct arithmetic compiler might avoid the two-way machine, folded controller, and local-window expansion. It must still encode unbounded finite derivations with a fixed number of unknowns; listing local inference polynomials alone does not solve that problem.

4. **Optimize with coefficient height charged.** The present 70-operation point treats constants as supplied. A useful second frontier would measure the bit length or arithmetic description length of every coefficient and the computational cost of producing it. The best construction under that metric could be a much larger polynomial with much smaller coefficients.

5. **Reduce unknowns without losing the fixed marker.** Sixteen is not a variable-minimality theorem. A sound elimination of a scale, ratio slack, or auxiliary coordinate could improve this instance, but any signed intermediate must be shown positive where the Pell arguments require it.

6. **Improve the degree/operation tradeoff.** The points $(70, 169)$, $(71, 156)$, and the transparent $(84, 56)$ describe different advantages. Can the fixed-input marker and the lower-degree auxiliary coefficient be combined with a cheaper single-polynomial finalizer?

7. **Obtain a weaker arithmetic internalization.** The statement proved here is formalizable in ZFC. A separately formalized internal PA derivation would require attention to the inherited K215 calibration as well as to the polynomial compiler. The repository explicitly leaves its internal PA endpoint distinct from its external proof-assistant theorem.

A Literal straight-line evaluator

The following table is generated from the delivered `fixed70` packet. The names `Bm1`, `Kconstant`, `MC`, `MF`, `W0` denote β, K, M, N, W, respectively. In particular `MF` is already shifted. Every row is one charged binary operation. Register names are implementation names; the mathematical definitions are in Section 2.

Table 1: The complete 70-operation evaluator.

Row	Output register	Binary expression
1	`C`	`Z` + `W0`
2	`q_minus_F`	`C` + `alpha`
3	`q`	`q_minus_F` + `F`
4	`q_minus_FZ`	`q_minus_F` − `Z`
5	`base_multiple`	`Bm1` × `Jrep`
6	`norm_repunit`	`q` − `base_multiple`
7	`tau_square`	`tau_root` × `tau_root`
8	`Lbig`	`q` × `q`
9	`n2`	`Lbig` × `q`
10	`wn2`	`w` × `Lbig`
11	`sn2`	`s` × `n2`
12	`UM`	`wn2` × `sn2`
13	`R10b`	`eta` + `zeta`
14	`ksn2`	`R10b` × `sn2`
15	`first_root_base`	`UM` × `ksn2`
16	`first_next`	`first_root_base` + `R10b`
17	`first_product`	`first_root_base` × `first_next`

Continued on the next page

Row	Output register	Binary expression
18	norm_first	tau_square − first_product
19	R10a	ksn2 + eta
20	R12	UM + sn2
21	cam2	R10a × R12
22	D1	wn2 + cam2
23	a4	4 × R12
24	a4m5	a4 + 3
25	gam	gamma × a4m5
26	R14	D1 + gam
27	L15	R14 × R14
28	a_square	R12 × R12
29	A	a_square + a4m5
30	c2	R10a × R10a
31	Ac2	A × c2
32	norm_main	L15 − Ac2
33	norm_pair	norm_first × norm_main
34	norm_triple	norm_pair × norm_repunit
35	aux_y2	y_aux × y_aux
36	hpm1	h × UM
37	index_difference	R10b − hpm1
38	gap_product	base_multiple × q_minus_F
39	gap	gap_product + q_minus_FZ
40	Lm1	Lbig − 1
41	rproduct	gap × Lm1
42	qMF	q × MF
43	mask_factor	MC + qMF
44	mask	mask_factor × Jrep
45	r_lhs	rproduct + mask
46	norm_index	index_difference − r_lhs
47	kinner	Kconstant + w
48	innerC	kinner × C
49	transport_partial	innerC + q_minus_F
50	local_rhs	transport_quotient × base_multiple
51	norm_transport	transport_partial − local_rhs
52	L16	f × f
53	auxiliary_Tf	auxiliary_quotient × f
54	auxiliary_Tf_minus_one	auxiliary_Tf − 1
55	auxiliary_c_Tf	R10a × auxiliary_Tf_minus_one
56	auxiliary_R_f2	r_lhs × L16
57	aux_u_rhs	auxiliary_c_Tf − auxiliary_R_f2
58	H2	aux_u_rhs × aux_u_rhs
59	aux_square_gap	H2 − aux_y2
60	aux_coefficient_root	i × Ac2
61	R16	aux_coefficient_root × aux_coefficient_root
62	scaled_f_square	A × L16
63	norm_strong	scaled_f_square − R16
64	L17	R16 × aux_square_gap
65	norm_aux	L17 + aux_y2
66	norm_four	norm_triple × norm_aux
67	norm_product	norm_four × norm_index
68	all_units	norm_product × norm_transport
69	seven_units	all_units × norm_strong
70	polynomial	seven_units − A

B Complete finite recipe for the five coefficients

This appendix specifies all five fixed integers in the main polynomial. Its powers, finite sums, and finite case distinctions concern only fixed coefficients: none contains a Diophantine unknown. The complete window list and the binary expansions of the resulting integers are not materialized. The correction bit defined below is likewise retained as an exact finite arithmetic expression, rather than claimed to have been numerically evaluated.

The literal K215 table and its normalized transition function are fixed in Section 9.1. For this appendix, denote that normalized function by

$$\delta(q,s)=(q',s',D),\qquad q,q'\in\{0,\dots,434\},\quad s,s'\in\{0,\dots,9\},\quad D\in\{-1,0,1\}.$$

Its start state is 0, its erase state is 1, its absorbing halt is 4, and its blank symbol is 0. In particular $\delta(0,1)=(1,2,0)$. The archive supplies the entire 4,350-row table, so this use of δ does not introduce an unspecified universal-machine index.

B.1 The complete ordered tile alphabet and local predicate

Write $\perp$ for an absent payload, phase, or head, as appropriate. A tile has four fields (v,h,π,φ): two boundary bits, a payload, and a phase. A payload is (s,q), with a tape symbol s and a state q or $\perp$. The ordered tile alphabet is as follows:

1. Tile 0 is $\mathsf{V}=(1,0,\perp,\perp)$ and tile 1 is $\mathsf{H}=(0,1,\perp,\perp)$.
2. For $s=0,\dots,9$ and $q=\perp,0,\dots,434$, in this order, include $(0,0,(s,q),\perp)$. Its tile index is
$$2+436s+\eta(q),\qquad \eta(\perp)=0,\quad \eta(q)=q+1\quad (q\neq\perp).$$
3. Tiles $4362,4363,4364,4365$ have phases $\mathsf{L},\mathsf{I},\mathsf{Q},\mathsf{R}$, respectively. Their boundary bits are zero, and their respective payloads are
$$(0,\perp),\quad (1,\perp),\quad (1,0),\quad (0,\perp).$$

Thus the alphabet has $a_c=4366$ elements. All the fields of every tile are fixed finite data.

Here is the complete cross predicate used to select windows. Let t,t_ℓ,t_r,t_d,t_u be the center, left, right, down, and up tiles. Subscripts on fields mean that they are read from the corresponding tile. First impose

$$v=v_u,\qquad vv_r=0,\qquad hh_u=0,\qquad (v=v_r=0\implies h=h_r). \tag{28}$$

If the center is V or H, these are its entire tests. For a nonboundary center, impose all the following additional conditions.

1. If $h_d=1$, its phase belongs to $\{\mathsf{L},\mathsf{I},\mathsf{Q},\mathsf{R}\}$ and its payload is the one prescribed above. A left V requires phase L; a right V requires phase R. If the right neighbor is not V, the two phases must form one of the six pairs
$$(\mathsf{L},\mathsf{L}),\ (\mathsf{L},\mathsf{I}),\ (\mathsf{I},\mathsf{I}),\ (\mathsf{I},\mathsf{Q}),\ (\mathsf{Q},\mathsf{R}),\ (\mathsf{R},\mathsf{R}).$$
If $h_d=0$, the center phase must instead be $\perp$.
2. If the center carries a head, its transition may not move left through a left V or right through a right V. If $h_u=1$, that head, if present, must be in halt state 4.

3. If $h_u = 0$, each non-V left or right neighbor must have a payload. Treat a neighboring V as payload $(0, \perp)$. The next center symbol is its current symbol when headless, and its written symbol under δ otherwise. Form the list of incoming states from the left head moving right, the center head staying, and the right head moving left. There must be at most one such state. The up payload must be the next center symbol together with that state, or with $\perp$ if the list is empty.

This is the literal predicate in the standalone `artifacts/compiler/finite_window_constants.py`; it retains every initialization, collision, no-escape, and final-row test of the inherited helical relation [10].

A 3×3 window is allowed exactly when its central cross satisfies these tests. Its four corner entries are unrestricted. Write entries in row-major order, with row 0 supplying the down neighbor and row 2 supplying the up neighbor. The two distinguished windows, expressed entirely in tile indices, are

$$\mathsf{S} = \begin{pmatrix} 1 & 1 & 1 \\ 4363 & 4363 & 4364 \\ 438 & 438 & 876 \end{pmatrix}, \qquad \mathsf{E} = \begin{pmatrix} 1 & 1 & 1 \\ 4362 & 4363 & 4363 \\ 2 & 438 & 438 \end{pmatrix}.$$

Both pass the predicate; they are the Start and End windows. The ordered list $w_0, \ldots, w_{k_c-1}$ begins with S, E and then contains every other allowed nine-tuple in lexicographic tile-index order. This defines a terminating enumeration with at most 4366^9 predicate tests, independent of whether the machine halts. Its exact size is

$$k_c = 113175624520853754856480928432. \tag{29}$$

The exact compressed census is the sum of 311471256024587 allowed central crosses, each with 4366^4 choices of corners. It is a finite partition count, not a sample estimate.

B.2 Evaluated fixed layout parameters

The following parameters determine the native positions, clause bands, and radices. The subscript c distinguishes compiler parameters from the Pell variables in the main polynomial.

Parameter	Fixed value
a_c	4366
ℓ_c	97
p_c	565878122604268774282404182 74
m_c	113175624520853754856480967730
U_c	113175624520853754856480993923
H_c	577195685056354149768053070 3172
$T_{c,1}$	118834405746896442599305043 79381
$T_{c,2}$	179949242988157470221804780 51224
g_c	210506661608787984033054648 87146

Put

$$A_c = 2^{\ell_c}, \qquad b = 5^{44}, \qquad L_c = 5^{45}, \qquad d = bL_c = 5^{89}, \qquad R_0 = 2^b, \qquad B = R_0^{L_c} = 2^d. \tag{30}$$

These values reproduce the deterministic layout rule:

$$\begin{aligned}
\ell_c &= \max\{2, \mathrm{bitlength}(k_c+1)\}, & m_0 &= 2\ell_c + 30a_c + 8,\\
p_c &= \max\left\{0, \left\lceil \frac{k_c + 9a_c + 4 - m_0}{2} \right\rceil \right\}, & m_c &= m_0 + 2p_c,\\
U_c &= m_c + 6a_c - 3, & H_c &= 51U_c + 3a_c + 1,\\
T_{c,1} &= 105U_c + 4a_c + 2, & T_{c,2} &= 159U_c + 4a_c + 3,\\
g_c &= 186U_c + 4a_c + 4.
\end{aligned}$$

The least power of five at least $213U_c + 4a_c + 5$ is L_c. The radix choice is justified after the clause coefficients are defined.

B.3 Native positions and all clause coefficients

For $r, s \in \{0,1,2\}$ and $u \in \{0, \dots, a_c - 1\}$, define

$$j_c(r,s,u) = 1 + 3a_c r + a_c s + u, \tag{31}$$

$$e_c(r,s,u) = k_c + 3a_c + (2-r)3a_c + (2-s)a_c + u. \tag{32}$$

There is exactly one ignored dummy position,

$$e_* = k_c + 12a_c = 113175624520853754856480980824.$$

The complete finite native position set is

$$\begin{aligned}
\mathcal{E}_c = {} & \{0, \dots, k_c - 1\} \cup \{e_c(r,s,u) : 0 \le r, s \le 2,\ 0 \le u < a_c\}\\
& \cup \{e_*\} \cup \{U_c, 3U_c, 9U_c, 27U_c\}.
\end{aligned} \tag{33}$$

Its members are distinct, and $|\mathcal{E}_c| = m_c + 1$.

Define a nonnegative fixed coefficient c_e at each position. For a selector position $0 \le i < k_c$, set

$$c_i = 1 + \sum_{r=0}^{2}\sum_{s=0}^{2} A_c^{j_c(r,s,(w_i)_{3r+s})} + \sum_{h=1}^{4} A_c^{9a_c+h}. \tag{34}$$

Here $(w_i)_t$ is the tile index at entry t of the already specified fixed window w_i. At the remaining positions put

$$\begin{aligned}
c_{e_c(r,s,u)} &= A_c^{j_c(r,s,u)}, & c_{e_*} &= 0,\\
c_{U_c} &= A_c^{9a_c+1}, & c_{3U_c} &= A_c^{9a_c+2},\\
c_{9U_c} &= A_c^{9a_c+3}, & c_{27U_c} &= A_c^{9a_c+4}.
\end{aligned} \tag{35}$$

The clause mask, including its fixed zero-expression padding clauses, is

$$\mu_c = A_c - 2 + \sum_{j=1}^{9a_c+4+p_c} A_c^j. \tag{36}$$

Omitting the padding summands would change the mask population and would not be the inherited compiler instance.

For clarity, the least-power-of-five radix rule can now be stated exactly. Let $C_\Sigma = \sum_{e\in\mathcal{E}_c} c_e$. Its target is

$$T_{\mathrm{radix}} = \max\{4(m_c+3)(2C_\Sigma+6)+8,\ 2\mu_c+4,\ 16\}.$$

Choose b to be the least power of five at least bitlength$(T_{\mathrm{radix}}-1)$. For this instance,

$$C_\Sigma < (14k_c+9a_c+5)A_c^{9a_c+4} < 2^{3812007},$$

so the first member of the maximum is below $2^{3812108}$. The second member dominates. Its exact bit-length contribution is

$$\begin{aligned}\mathrm{bitlength}(2\mu_c+3) &= \ell_c(9a_c+4+p_c)+2\\ &= 548901778926140711053932438 4486.\end{aligned}$$

Its least power-of-five majorant is 5^{44}, as asserted in (30). No giant power has to be expanded to verify these inequalities.

B.4 Convolution coefficients and the fixed correction bit

The uncorrected center coefficient and the right coefficient are

$$\begin{aligned}D_{C,0} &= R_0^{3a_c} + R_0^{H_c+a_c} + R_0^{8U_c} + R_0^{24U_c}\\ &\quad + \sum_{e\in\mathcal{E}_c} c_e\big(R_0^{T_{c,1}-e} + R_0^{T_{c,2}-e}\big),\end{aligned} \tag{37}$$

$$D_R = R_0^{H_c}. \tag{38}$$

All exponents are nonnegative fixed integers. Define the single fixed bit

$$\epsilon_c = \begin{cases}1, & 2D_{C,0}-D_R \equiv 0 \pmod 5,\\ 0, & 2D_{C,0}-D_R \not\equiv 0 \pmod 5,\end{cases} \qquad D_C = D_{C,0} + \epsilon_c R_0^{g_c}. \tag{39}$$

This is a finite arithmetic decision on uniquely defined fixed integers. It is not a Diophantine variable, an unspecified choice of coefficient, or a query about consistency. Since b is a power of five, $R_0 \equiv 2 \pmod 5$. The added monomial has nonzero residue after multiplication by 2, so the definition guarantees

$$2D_C - D_R \not\equiv 0 \pmod 5.$$

It can in principle be evaluated using modular exponentiation and the fixed finite sums. This packet does not claim to have evaluated ϵ_c or the window-dependent sum in (37).

B.5 Native masks, shifted field mask, and final ports

Define the old field mask by the exact finite expression

$$\begin{aligned}M_{F,0} &= \mu_c\big(R_0^{T_{c,1}} + R_0^{T_{c,2}}\big)\\ &\quad + \sum_{r=0}^{1}\sum_{s=0}^{2}\sum_{u=0}^{a_c-1} R_0^{e_c(r,s,u)}\\ &\quad + \sum_{r=0}^{2}\sum_{s=0}^{1}\sum_{u=0}^{a_c-1} R_0^{H_c+e_c(r,s,u)} + R_0^{9U_c} + R_0^{27U_c}.\end{aligned} \tag{40}$$

The modified half-binomial masks are

$$M_C = B - 1 - \sum_{\substack{e \in \mathcal{E}_c \\ e \neq 1}} R_0^e - 2R_0^{e_*}, \tag{41}$$

$$M_F = M_{F,0} + 4. \tag{42}$$

The first modification frees the second bit at the ignored dummy position; the second adds the low verification bit. These are the modifications of the reviewed compiler export [10]. In particular

$$0 < M_C, M_F < B - 1, \qquad M_C \equiv 2 \pmod 4, \qquad M_F \equiv 4 \pmod 8,$$
$$\operatorname{pop}(M_C) + \operatorname{pop}(M_F) = d.$$

The five coefficients supplied to the main polynomial are now exactly

$$\boxed{\begin{aligned} &\beta = B - 1, && K = D_C + BD_R, \\ &M = M_C, && N = M_F + B - 1, \\ &W = R_0 B^2 = 2^{2 \cdot 5^{89} + 5^{44}}. \end{aligned}} \tag{43}$$

The source coefficient N is already shifted. It is *not* the native one-cell mask M_F; replacing one by the other changes the polynomial. The endpoint W inserts the End selector at cyclic cell 2, giving the three-I initialization run and four unary ones already proved equivalent to blank K215.

B.6 Explicitness and the representation boundary

Every summation bound, finite set, field lookup, and exponent in this appendix is fixed before the sixteen unknowns of the main polynomial are chosen. Substitution of (43) therefore produces one ordinary integer-coefficient polynomial. No variable power, bounded universal quantifier, sequence parameter, or acceptance test remains in that polynomial.

The compressed representation is essential for practical delivery: B alone has $5^{89} + 1$ binary digits. This coefficient height is not included in the 70-operation evaluation count. The count assumes access to the five fixed numerals, as stated in the main theorem; it does not assert that their construction or storage is inexpensive. The companion `artifacts/compiler/COEFFICIENTS.md`, source code, and pinned manifests give the same recipe in machine-readable and prose forms.

All compiler formulas in this appendix are the lazy form of the retained construction

`new_constants(compile_windows(windows,a)).`

The new explicit controller instantiation does not replace the inherited proof of that compiler, and the exact finite specification must not be confused with a claim that all its giant integers have been expanded.

C Source and dependency references

The source pin for every ProveIt item below is `cdd582bec13f908df3711fa9e05d1ddac368a5db`. The delivered provenance manifest records full paths and byte hashes for the source artifacts used. These references distinguish inherited theorems from the new fixed-input reductions.

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